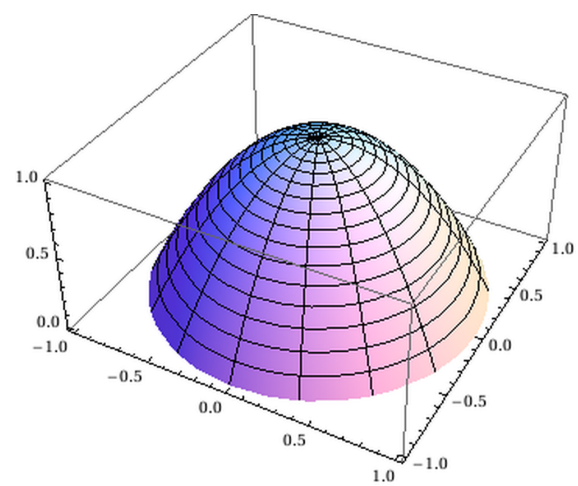
Worksheet 11 solution - Surface integrals of vector fields and related theorems

1. Consider the region D in $\mathbb{R}^3$ bounded by the xy-plane and the surface $x^2 + y^2 + z = 1$.

   (a) Make a sketch of D.

   Solution. The sketch of D is shown below.



   (b) The boundary of D, denoted ∂D, has two parts: the curved top S_1 and the flat bottom S_2. Parameterize S_1 and calculate the flux of $\mathbf{F} = (0, 0, z)$ through S_1 with respect to the upward pointing unit normal vector field. Check you answer with the instructor.

   Solution. To parametrize S_1, one has

$$\mathbf{r}(u,\ v) = \langle u\cos v,\ u\sin v,\ 1 - u^2\rangle, \quad 0 \le u \le 1,\ 0 \le v \le 2\pi.$$

   In order to calculate the flux, first we have

$$\mathbf{r}_u = \langle \cos v,\ \sin v,\ -2u\rangle, \quad \mathbf{r}_v = \langle -u\sin v,\ u\cos v,\ 0\rangle,$$

   and so

$$\mathbf{r}_u \times \mathbf{r}_v = \langle 2u^2\cos v,\ 2u^2\sin v,\ u\rangle.$$

   Therefore, the flux of $\mathbf{F} = (0, 0, z)$ through S_1 with respect to the upward pointing unit normal vector field is

$$\begin{aligned}
\iint_{S_1} \mathbf{F}\cdot\mathbf{n}\, dS &= \int_0^{2\pi}\int_0^1 \mathbf{F}(u,v)\cdot(\mathbf{r}_u \times \mathbf{r}_v)\, du\, dv \\
&= \int_0^{2\pi}\int_0^1 (0,0,1-u^2)\cdot\langle 2u^2\cos v,\ 2u^2\sin v,\ u\rangle\, du\, dv \\
&= \int_0^{2\pi}\int_0^1 (1-u^2)u\, du\, dv = \frac{\pi}{2}.
\end{aligned}$$

(c) Without doing the full calculation, determine the flux of $\mathbf{F}$ through S_2 with the downward pointing normals.

Solution. Since $\mathbf{F} = 0$ on S_2, we know the flux of $\mathbf{F}$ through S_2 is

$$\iint_{S_2} \mathbf{F} \cdot \mathbf{n}\, dS = \iint_{S_2} \mathbf{0} \cdot \mathbf{n}\, dS = 0.$$

(d) Determine the flux of $\mathbf{F}$ through ∂D with the outward pointing normals.

Solution. By adding up the result from (a) and (b), one gets

$$\iint_{\partial D} \mathbf{F} \cdot \mathbf{n}\, dS = \iint_{S_1} \mathbf{F} \cdot \mathbf{n}\, dS + \iint_{S_2} \mathbf{F} \cdot \mathbf{n}\, dS = \frac{\pi}{2}.$$

(e) Apply the Divergence Theorem and your answer in (d) to find the volume of D. Check your answer with the instructor.

Solution. By the Divergence Theorem, one has

$$\iint_{\partial D} \mathbf{F} \cdot \mathbf{n}\, dS = \iiint_D \operatorname{div}\mathbf{F}\, dV.$$

Since $\operatorname{div}\mathbf{F} = 1$, one gets

$$\text{Volume}(D) = \iiint_D 1\, dV = \iiint_D \operatorname{div}\mathbf{F}\, dV = \iint_{\partial D} \mathbf{F} \cdot \mathbf{n}\, dS = \frac{\pi}{2}.$$

2. Consider the vector field $\mathbf{F} = (-y, x, z)$.

(a) Compute curl $\mathbf{F}$.

Solution.

$$\text{curl } \mathbf{F} = \begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -y & x & z \end{pmatrix} = (0, 0, 2).$$

(b) For the surface S_1 above, evaluate $\iint_{S_1} (\text{curl } \mathbf{F}) \cdot \mathbf{n}\, dA$.

Solution. By applying the parametrization of S_1 in 1(b), one gets

$$\begin{aligned} \iint_{S_1} (\text{curl } \mathbf{F}) \cdot \mathbf{n}\, dA &= \int_0^{2\pi} \int_0^1 (\text{curl } \mathbf{F}) \cdot (\mathbf{r}_u \times \mathbf{r}_v)\, du\, dv \\ &= \int_0^{2\pi} \int_0^1 2u\, du\, dv = 2\pi. \end{aligned}$$

(c) Check your answer in (b) using Stokes' Theorem.

Solution. The boundary of S_1 is a unit circle centered at the origin in the xy-plane. So we can parametrize it as

$$C: \ \mathbf{r}(t) = \langle \cos t, \ \sin t, \ 0 \rangle, \quad 0 \le t \le 2\pi.$$

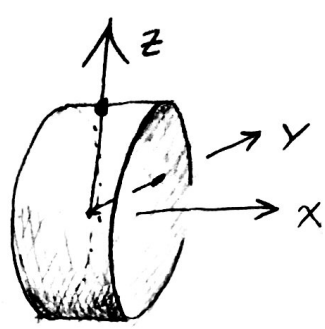
Thus, by Stokes' Theorem, one has

$$\iint_{S_1} (\operatorname{curl} \mathbf{F}) \cdot \mathbf{n} \, dA = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \, dt$$
$$= \int_0^{2\pi} (-\sin t, \ \cos t, \ 0) \cdot \langle -\sin t, \ \cos t, \ 0 \rangle \, dt = \int_0^{2\pi} 1 \, dt = 2\pi.$$

3. Let S be the portion of the cylinder of radius 2 about the x-axis where $-1 \le x \le 1$.

(a) Draw a picture of S and compute its area without doing any integrals. Hint: How could you make this cylinder out of paper?

SOLUTION: We have Area$(S) = 8\pi$ since we can cut S along a line in the xy-plane and flatten it out into a $2 \times 4\pi$ rectangle.



(b) Find a parameterization $\mathbf{r}(u, v)$ of S.

SOLUTION:

Let $u = x$ and $v =$ angle about the $x-$axis, so that

$$r(u, v) = (u, 2\cos v, 2\sin v)$$

with domain $D = \{-1 \le u \le 1 \text{ and } 0 \le v \le 2\pi\}$.

(c) Does the normal vector field associated to your parameterization point into or out of S? First, try to determine this without doing any calculations, and then check your answer by evaluating $\mathbf{r}_u \times \mathbf{r}_v$.

SOLUTION:

By the right hand rule, the vector points inwards. Check:

$$\mathbf{r}_u \times \mathbf{r}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 0 \\ 0 & -2\sin v & 2\cos v \end{vmatrix} = \langle 0, -2\cos v, -2\sin v \rangle$$

At $u = 0, v = 0$ we have $\mathbf{r}_u \times \mathbf{r}_v = \langle 0, -2, 0 \rangle$ which points inwards.

(d) If necessary, change your parameterization so that the normal vector field points *inwards*.

SOLUTION:

If your vector points outwards, interchanging the role of u and v will reverse its direction.

(e) Now consider the vector field $\mathbf{F} = \langle -z,\ xz,\ -xy \rangle$. Compute curl $\mathbf{F}$.

SOLUTION:

$\text{curl}\mathbf{F} = \langle -2x, y-1, z \rangle$

(f) Check that curl $\mathbf{F}$ is the sum of $\mathbf{G} = \langle -2x,\ -1,\ 0 \rangle$ and $\mathbf{H} = \langle 0,\ y,\ z \rangle$.

SOLUTION:

$\mathbf{G} + \mathbf{H} = \langle -2x, -1, 0 \rangle + \langle 0, y, z \rangle = \langle -2x, y-1, z \rangle = \text{curl}\mathbf{F}$

(g) Use geometric arguments to determine whether the flux of $\mathbf{G}$ is positive, zero, or negative. Remember that we have oriented S so that the normals point inwards. Do the same for $\mathbf{H}$ and curl $\mathbf{F}$.

SOLUTION:

Since every normal vector to S has 0 x-component, the flux of $\mathbf{G}$ is the same as the flux of $\langle 0, -1, 0 \rangle$, which by symmetry must be equal to 0. For $\mathbf{H}$, at each point of S $\mathbf{H}$ points outward from the surface, so the flux is negative (since the normals point inwards). In fact, $\mathbf{H} = -2\mathbf{n}$.

(h) Using your parametrization, directly compute the flux of curl $\mathbf{F}$.

SOLUTION:

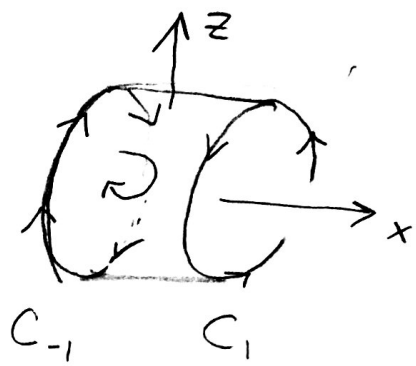
$$\int\int_S \text{curl}\mathbf{F} \cdot \mathbf{n} dA = \int_{-1}^{1}\int_0^{2\pi} \langle -2u, 2\cos v - 1, 2\sin v \rangle \cdot (\mathbf{r}_u \times \mathbf{r}_v) dv du$$

$$= \int_{-1}^{1}\int_0^{2\pi} 4(-\cos^2 v + \cos v + \sin^2 v) dv du = 4\int_{-1}^{1}\int_0^{2\pi} -1 + \cos v dv du = 4\int_{-1}^{1} -2\pi du = -16\pi$$

(i) Check your answer in (h) using Stokes' Theorem. Note here that ∂S has two boundary components, and make sure that your orient them correctly.

SOLUTION:

The boundary components should be oriented like so:



Parametrize C_1 by $\langle 1, 2\cos t, 2\sin t\rangle$ for $0 \le t \le 2\pi$ and C_2 by $\langle -1, 2\cos t, -2\sin t\rangle$ for $0 \le t \le 2\pi$. Now

$$\int_{C_1} \mathbf{F}\cdot d\mathbf{r} = \int_0^{2\pi} \langle -2\sin t, 2\sin t, -2\cos t\rangle \cdot \langle 0, -2\sin t, 2\cos t\rangle \, dt = \int_0^{2\pi} -4(\sin t^2 t + \cos^2 t)\, dt = -8\pi.$$

Similarly, $\int_{C_2} \mathbf{F}\cdot d\mathbf{r} = -8\pi$. Thus $\int_{\partial S} \mathbf{F}\cdot d\mathbf{r} = -8\pi + (-8\pi) = -16\pi = \int\int_S (\text{curl}\mathbf{F})\cdot \mathbf{n}\, dA$ as guaranteed by Stokes' theorem.

(j) Check your answer in (h) a second time by using what you learned in (g) to compute the flux of **G** and **H**.

SOLUTION:

We know curl$\mathbf{F} = \mathbf{G} + \mathbf{H}$ and the flux of **G** through S is 0. Since $\mathbf{H} = -2\mathbf{n}$, $\mathbf{H}\cdot\mathbf{n} = -2$. Hence

$$\int\int_S (\text{curl}\mathbf{F})\cdot\mathbf{n}\, dA = \int\int_S \mathbf{H}\cdot\mathbf{n}\, dA = \int\int_S -2\, dA = -2\text{Area}(S) = -2\cdot 8\pi = -16\pi$$